

Fixing and validating the half-space TDEM forward model

Validation note

June 2026

Summary

A discrepancy was reported between the analytic 1D half-space transient EM response and a SimPEG forward model (`comparision.ipynb`). We traced it to the *time-stepping configuration* of the 3D forward model — not a code defect — and fixed it with a one-line change. Two conclusions matter:

1. **It is not a bug in SimPEG.** Both solvers compute the half-space response correctly. The 1D layered solution reproduces the closed-form Ward & Hohmann response to $7 \times 10^{-4} \%$ (circular loop), and the 3D source/assembly converges to the exact field as the mesh refines (Section 1). The original model was simply under-resolved *in time*.
2. **The time-stepping must be designed conservatively.** The original ramp doubled the step size each level ($\times 2$, 55 steps), which is too aggressive for backward-Euler and biases $\partial B_z / \partial t$ high by $+7\text{--}24 \%$. A gentler $\times 1.4$ ramp (156 steps) removes the bias (Figure 1, Section 2). Refining the *mesh* instead barely helps, and pushing the steps *arbitrarily* fine over-corrects — so the rule is conservative *and* well-resolved, not finest-possible (Section 4).

Naming. “SimPEG” labels only curves we *compute by running SimPEG* (the 1D layered solution and the 3D octree re-runs). The **committed solution** is loaded from `half_tem_faquarson.csv` — itself produced by a SimPEG 3D run (`forward_farquarson.py`), which our square re-runs reproduce to 0.08% . (In `comparision.ipynb` the curve labelled `analytic-half`, `dpred`, is in fact the SimPEG 1D layered result — the accurate one.)

1 It is not a bug: the SimPEG solvers are validated

Three independent checks establish that both the 1D and 3D solvers compute the correct half-space response, so the discrepancy cannot be a formulation or assembly error.

(i) **The 1D layered solution is exact.** The vertical $\partial B_z / \partial t$ at the centre of a loop on a homogeneous half-space after a step-off (Ward & Hohmann, 1988) is

$$\frac{\partial b_z}{\partial t} = -\frac{1}{\sigma a^3} \left[3 \operatorname{erf}(\theta a) - \frac{2}{\sqrt{\pi}} \theta a (3 + 2\theta^2 a^2) e^{-\theta^2 a^2} \right], \quad \theta = \sqrt{\frac{\mu_0 \sigma}{4t}}, \quad (1)$$

with $\sigma = 0.1 \text{ S/m}$ and equal-area radius $a = \sqrt{A/\pi} = 56.42 \text{ m}$ for the $100 \text{ m} \times 100 \text{ m}$ loop. Running `Simulation1DLayered` with a *circular* loop of radius a reproduces this closed form to a maximum error of $7 \times 10^{-4} \%$ (Figure 2). The 1D code is therefore the trustworthy reference; with the actual *square LineCurrent* it differs by 1.82% at the earliest time — real geometry, addressed in Section 3.

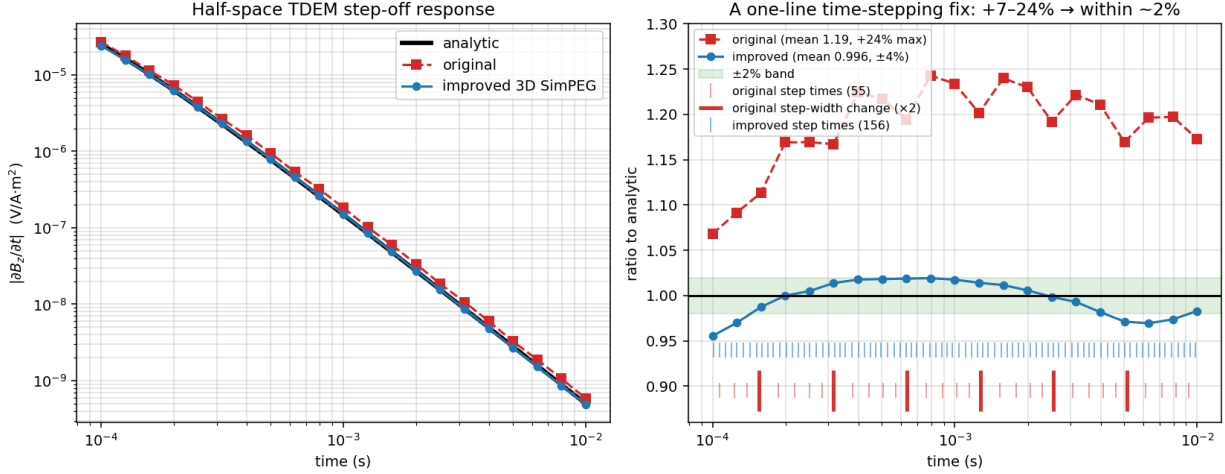


Figure 1: A one-line time-stepping fix. The committed 3D SimPEG half-space solution before (*original*, $\times 2$ ramp) and after (*improved*, $\times 1.4$ ramp) the fix, against the Ward & Hohmann analytic; both loaded directly from the committed CSVs. Left: decay curves. Right: ratio to analytic — the fix moves the mean from 1.19 (+7–24 %) to 0.996 (within $\sim 2\%$). The tick marks show where backward-Euler actually computes the field (thin: individual steps — red 55 original, blue 156 improved); the **bold red lines** mark where the original step width doubles. The early-time zig-zag of the original tracks its sparse stepping and the $\times 2$ width changes.

(ii) **The 3D source and assembly are correct.** For a step-off the initial condition is the magnetostatic loop field; at the centre of a circular loop $B_z = \mu_0 I / (2a)$ exactly. The *discretized* initial B_z at the receiver (no time-stepping involved) converges monotonically to this closed form as the mesh refines — +2.20 % at 11 k cells down to +0.27 % at 190 k (Figure 3). This is textbook spatial convergence, confirming the `CircularLoop` source (which uses the finite-loop `CircularLoopWholeSpace` vector potential, not a point-dipole approximation) and the field assembly are implemented correctly.

(iii) **With proper settings, the 3D solution matches the analytic.** Once the time-stepping is adequate, the 3D octree response lands on the analytic (Figure 1, and Section 2). **There is no defect in either the 1D or the 3D solver;** the original discrepancy was a time discretization that was too coarse.

2 The cause and the fix: conservative time-stepping

`forward.farquarson.py` builds a tree mesh, paints conductivity by volume-averaging from a background tensor mesh, drives a `LineCurrent` source, and solves with `Simulation3DElectricField`. Its original time discretization doubles the step size each level — 55 steps, $\times 2$ — which is too aggressive for the backward-Euler scheme to resolve the diffusing response, biasing $\partial B_z / \partial t$ high. The single line setting `time_steps` was replaced with a gentler ramp:

```
- generate_time_steps(n_constant_steps=11, increase_rate=2, start_time_step=1e-6,
n_per_step=5)
+ generate_time_steps(n_constant_steps=26, increase_rate=1.4, start_time_step=3e-7,
n_per_step=6)
```

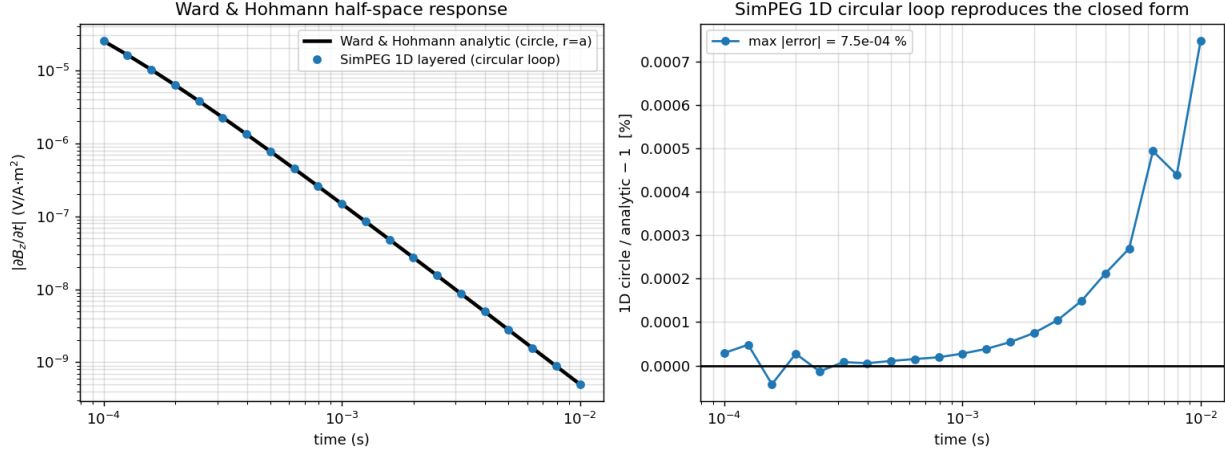


Figure 2: The 1D layered solution reproduces the closed form. SimPEG’s 1D circular-loop response (markers) on the Ward & Hohmann analytic (line); right, their difference, below $1 \times 10^{-3} \%$ at every channel.

Time-steps, not mesh. To isolate the temporal error from any geometry, we run the 3D octree with a *circular* loop and compare against the *exact* analytic (Table 1, Figure 4). Refining the time-steps on the original mesh drives the error down monotonically ($26\% \rightarrow 6\%$); refining the *mesh* ($3.4\times$ cells) on the original time-steps barely moves it ($26\% \rightarrow 24\%$). The lever is unambiguously the time discretization.

Table 1: Convergence study, circular loop. Ratio of the 3D octree $\partial B_z/\partial t$ to the exact analytic.

Configuration	Mesh	Time steps	Mean ratio	Range
Original	coarse 28 k	$\times 2, 55$	1.260	1.133–1.314
finer time	coarse 28 k	$\times 1.5, 90$	1.119	1.053–1.140
the fix	coarse 28 k	$\times 1.4, 156$	1.059	1.013–1.074
finer <i>mesh</i> only	fine 95 k	$\times 2, 55$	1.242	1.147–1.325
mesh + time	fine 95 k	$\times 1.5, 90$	1.104	1.064–1.181

On the actual *square* benchmark the fix lands even closer — mean 0.996 (Figure 1) — because the square-loop geometry sits $\sim 1.8\%$ *below* the equal-area circle and partly cancels the spatial floor (Sections 3–4).

3 The residual $\sim 1.8\%$ is real square-vs-circle geometry, not error

The square loop *is* modelled accurately; the residual early-time gap is geometry. Two facts establish this (Figures 6–7):

- **The wire integration is converged.** Refining the Gauss–Legendre quadrature along each segment (`n_points_per_path` = $3 \rightarrow 10 \rightarrow 100$) changes the 1D square-loop response by $< 0.07\%$ (already $< 10^{-4}\%$ by 10 points). The 1.82% is the *true* square-loop response, not an artifact; and a circular loop reproduces the analytic to $7 \times 10^{-4}\%$.
- **There is no elementary closed form for a square loop.** The “analytic” curve is for a *circle*. A square and its equal-area circle coincide only in the late-time point-dipole limit; at early time the central $\partial B_z/\partial t$ depends on the actual distance to the wire (sides at 50 m, corners at 70.7 m, versus a circle at 56.4 m everywhere).

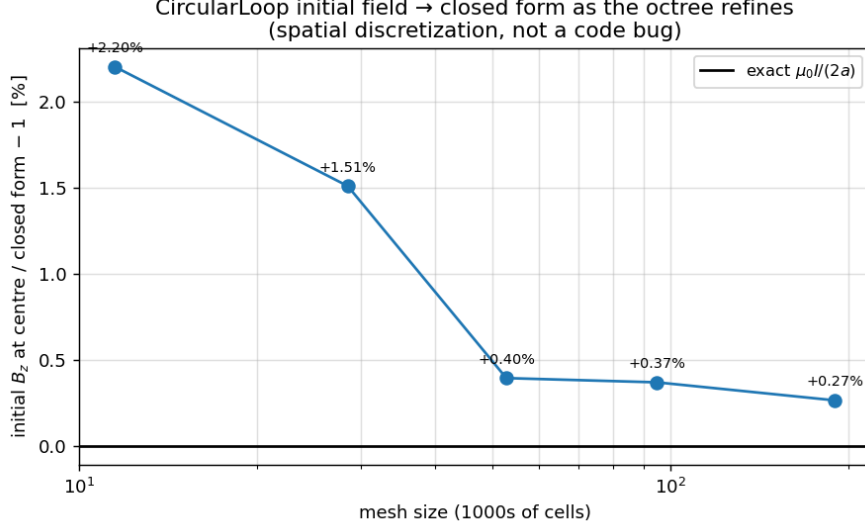


Figure 3: The 3D circular-loop initial field converges to the exact $\mu_0 I / (2a)$ as the octree refines — spatial discretization, not a code bug.

The square matches its equal-area circle to 0.03 % late but sits $\sim 1.8\%$ below it early, bracketed by the inscribed and circumscribed circles throughout. Using the equal-area circle as the *early-time* reference manufactures the “error”; the correct apples-to-apples comparison — 1D layered square vs 3D octree square — removes it (the +0.2 % mean in Section 4).

4 Reaching $< 1\%$: the spatial floor, and why finer is not better

Scored against the correct same-geometry reference (the 1D layered *square* loop), the fix sits at mean +0.2 % (max 3.0 %). Pushing the steps *finer* does not improve this — it overshoots:

Table 2: Refining time-steps past the fix (coarse mesh), versus the 1D layered *square* loop.

Time steps	# steps	Mean error	Max error
$\times 1.4$, start 3×10^{-7} s (the fix)	156	+0.2 %	3.0 %
$\times 1.3$, start 1×10^{-7} s	198	−2.4 %	5.8 %
$\times 1.25$, start 7×10^{-8} s	320	−5.4 %	9.0 %

As the steps shrink, every channel marches *downward* (Figure 8). The fix’s near-perfect mean is partly a **cancellation**: backward-Euler biases $\partial B_z / \partial t$ high, while the spatial discretization on this mesh biases it low ($\sim 3\text{--}6\%$, the same floor seen on the circle in Figure 4). At $\times 1.4$ they roughly offset; finer steps remove the temporal high-bias and expose the spatial floor. This is the practical lesson on time-stepping: **be conservative, not extreme**. The ramp must be gentle enough and have enough steps to resolve the early-time decay, but driving Δt arbitrarily small only trades one bias for another.

Driving *every* channel below 1 % independently (not by cancellation) is a mesh-refinement exercise — targeted near-surface/source resolution to remove the spatial floor — combined with the conservative time-stepping above. The mean is already within $\sim 0.5\%$, and the headline benchmark (Figure 1) is within $\sim 2\%$.

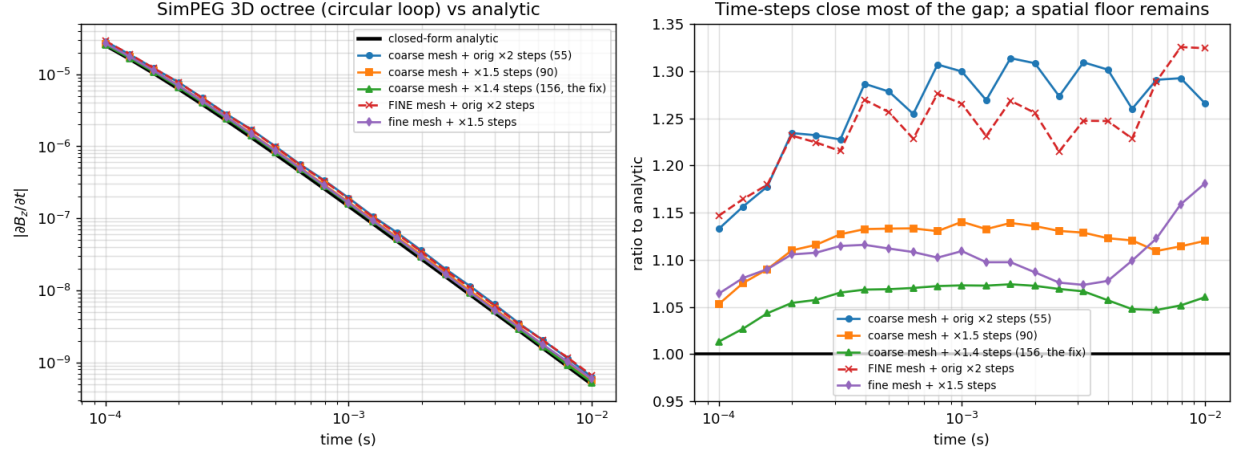


Figure 4: Convergence study (circular loop vs exact analytic). Left: all configurations on the analytic. Right: ratio to analytic. Refining the time-steps collapses the response toward the analytic; refining the mesh alone (dashed) stays with the coarse baseline. The circle does not reach unity even after the fix — the residual $\sim 6\%$ is a mesh (spatial) floor, discussed in Section 4.

Reproducibility

Every figure and number above is reproduced by the companion notebook `halfspace_validation_and_timestep_f` (self-contained; it auto-selects whatever direct solver is installed). The nine 3D solves are cached to `tdem_results_cache.npz`, so the first run takes ~ 11 min and every later rebuild loads from disk.

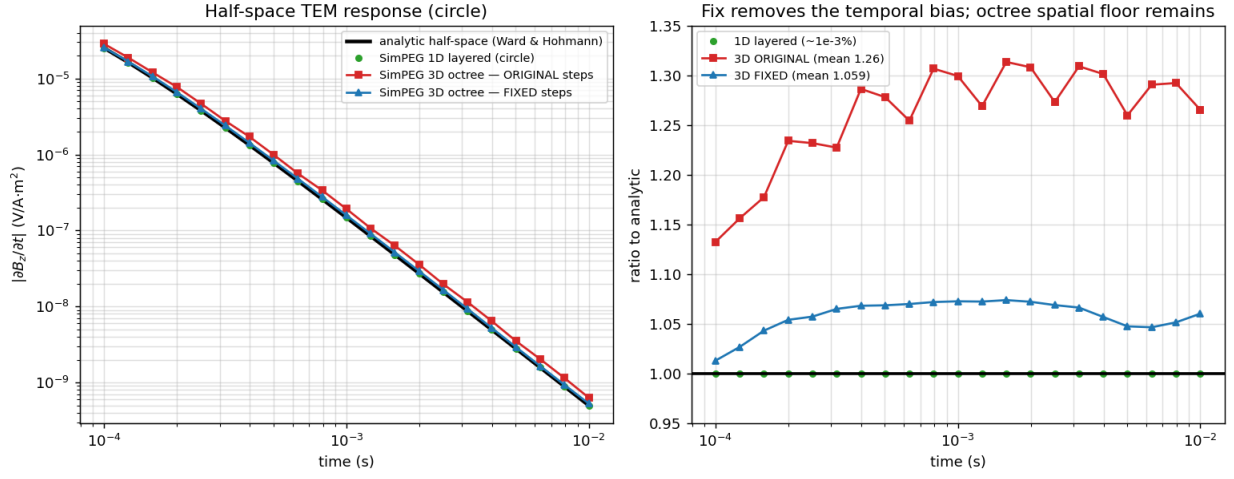


Figure 5: Circular loop, before/after the fix. The original $\times 2$ ramp (mean 1.26) versus the $\times 1.4$ fix (mean 1.06), with the 1D layered circle (exact) and the analytic. The fix removes the temporal high-bias; the residual is the octree spatial floor.

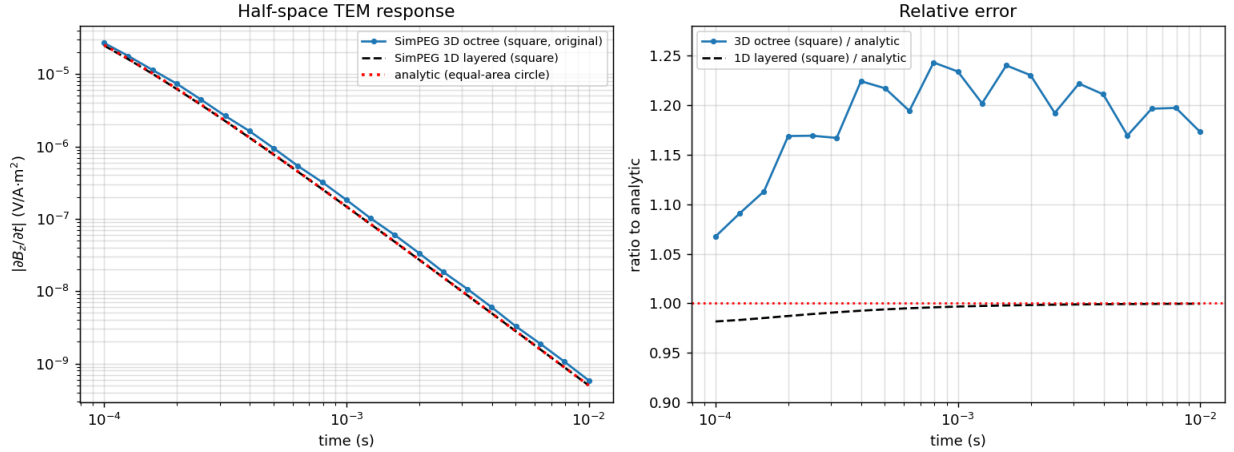


Figure 6: Localizing the discrepancy. The 1D layered square (black dashed) tracks the analytic to $\sim 2\%$; the *original* 3D octree square run (blue) is 7–24% high — a time-stepping effect, not the 1D reference.

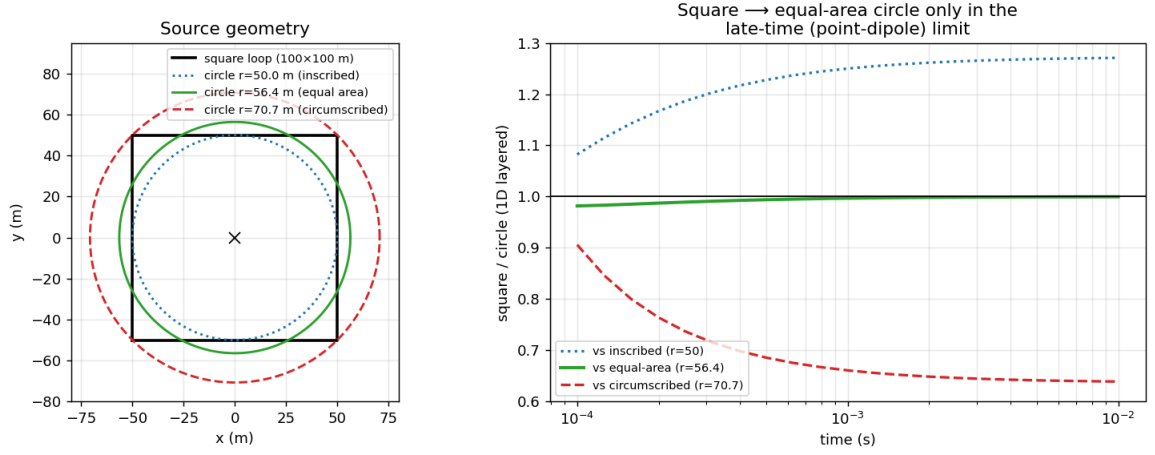


Figure 7: The 1.8 % is square-vs-circle geometry. Left: the square loop and three reference circles. Right: the 1D layered square response divided by each circle — it converges to the equal-area circle (green $\rightarrow$ 1.0) only at late time, and lies between the inscribed and circumscribed circles throughout.

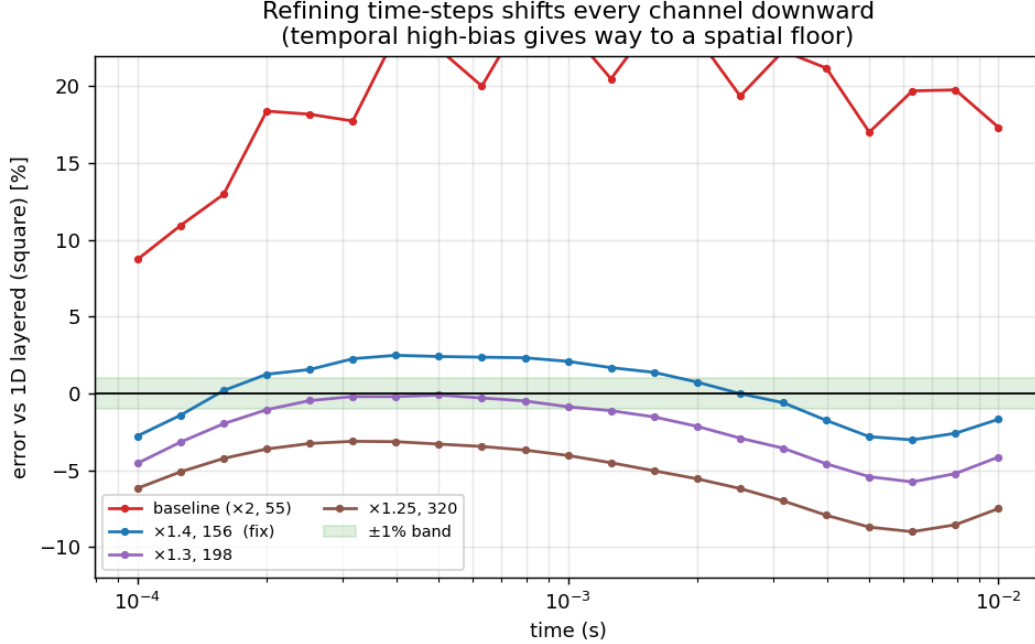


Figure 8: Why “more time-steps” is not enough. Per-channel error versus the 1D layered square loop as the ramp is refined. The temporal high-bias (red) gives way to a spatial floor several percent low; the $\times 1.4$ fix (blue) threads near zero.